

Victor

Full-stack engineer · Backend architecture · Systems programming · Modern web
github.com/Nyxox-debug

// Experience

Backend Engineer — HireusLogistic [Recent]

TypeScript · MongoDB · Redis · OAuth 2.0 · Docker · Kubernetes

- Scaled real-time rider tracking and dispatch system using WebSockets and Redis pub/sub
- Implemented OAuth 2.0 authentication and role-based access control across all API surfaces
- Designed MongoDB schemas optimized for geospatial queries and high-throughput rider matching
- Containerized services with Docker and orchestrated deployments with Kubernetes

Backend Engineer — ITSA [Hackathon]

TypeScript · Node.js · Backend Development · Testing

- Collaborated with a cross-functional team to improve system performance
- Diagnosed, reported, and resolved software issues
- Wrote, reviewed, and validated backend code and tests to ensure reliability

Frontend Engineer — Uniskill [Hackathon]

React · Next.js · TypeScript · Solana

- Built decentralized e-learning platform with blockchain-based certificate storage
- Integrated Solana smart contracts to store and verify course certificates on-chain
- Implemented Next.js App Router for SSR and static generation, optimizing performance
- Managed complex client-side state and REST API interactions for real-time tracking
- Ensured WCAG 2.1 accessibility compliance and responsive cross-browser functionality

Backend Developer — UNITE [Hackathon]

IPFS · Decentralized Storage · Node.js · TypeScript

- Designed and integrated IPFS-based storage for secure, distributed file management
- Developed backend APIs for seamless file upload, retrieval, and indexing
- Collaborated with team to optimize data consistency and storage efficiency

// Education

B.Sc. Computer Science FUTO [Ongoing]

Data Structures · Operating Systems · Computer Architecture · Compiler Design · Distributed Systems · Linear Algebra

// Selected Projects

Synap — Autograd / ML Framework [Recent]

C++, Python, pybind11

Minimal ML framework implementing tensors, reverse-mode autodiff, and MLP primitives from scratch. C++ core with Python bindings via pybind11. Zero-copy tensor views through shared storage.

3D Model Renderer [Recent]

C++, OpenGL, GLSL

Real-time renderer with OBJ loading, multi-texture support, custom GLSL shaders, and a first-person camera. Full MVP transform pipeline, delta-time movement, no game engine.

Bat — Interpreter from Scratch

Go

Complete interpreted language built from scratch — lexer, Pratt parser, AST evaluator, closures, and an interactive REPL. First-class functions with closures via environment capture.

GitHub Code Analyzer

Go, Python, tree-sitter

Clones any public GitHub repo and performs static analysis using ASTs across 25+ languages, extracting complexity metrics, structure, and dependencies.

Mini-Fig – Vector Editor

C++, TypeScript, Emscripten

Vector graphic editor and lightweight 2D drawing engine built with C++ and WebAssembly, backed by a React frontend.

// Hackathons

ITSA Hackathon

🏆 Grand Prize – Best Fintech Hack

Node.js · TypeScript · MongoDB

Competed with 15 teams to identify issues and propose improvements on a live production codebase.

Hack4Futo 2023

🏆 5th Place – 25+ teams

Python · TypeScript · React · OpenAI API · Rust · Solana

Built an AI-powered education platform that tracks learning progress and stores user certificates on-chain.